PART I THE FUNDAMENTALS OF FINANCE, MARKETS, VALUATION AND FINANCIAL FIRMS

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1 THE PRICE MECHANISM AND THE THREE PILLARS OF FINANCE

WHAT IS FINANCE?

If the 'man in the street' were asked 'What is finance?', some of the possible answers might be: 'It's to do with money'; 'It's about getting a mortgage'; 'It's what banks do'; 'It's about saving for retirement'. All of these answers would be correct. But what they all omit is two important words: **risk** and **price** . Finance is about assessing risk, pricing it and then transferring it through credit and capital markets. A new issue of shares, for example, is about pricing the riskiness of the company concerned and then transferring that risk to new investors through the capital markets. If that company subsequently goes to a bank to raise a loan for expansion, the bank manager's job is to assess the risk of a loan to that particular company, price that risk as an interest rate to be charged and then transfer that risk on to the bank's own balance sheet.

THE LIFECYCLE VIEW OF FINANCE

A more intuitive, and personal, understanding of the role of the finance industry would have as its starting point the **human economic lifecycle**. This starts with the observation that people's income is highly variable during their lifetime. In the early years of adulthood, students (or their parents) will have to find a means to finance their education at a time when their income is small or zero. Once employed their income would, in least in previous generations, increase year by year. But their income will fall to zero or close to zero if they lose their employment, if they are ill for a long period and, with certainty, when they retire from employment. If income falls short of essential spending on food and accommodation, an individual has a problem. To overcome this potential problem, people build up a 'cushion' of wealth by spending below their income when times are good, i.e. by **saving**. The greatest need for saving is to finance consumption in retirement a period when it is known, well in advance, that there will be no, or little, income generation. Saving for retirement involves regular month-by-month saving throughout working life in order to build up a large cash sum often known as a 'pension pot'.

It is these cushions of wealth that individuals build up which provide the raw material for the whole finance industry. They are built up, i.e. **accumulated**, during times when income exceeds spending and **decumulated** during periods when income falls short of day-to-day expenditure. We call this saving and dis-saving the **intertemporal transfer** of spending power, which simply

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means moving spending power backwards or forwards in time. Thus the finance industry enables each individual or household in the economy to **store surplus income as capital** during good times and **convert capital back into income**, i.e. 'releasing' it again, during bad times. In aggregate across all individuals in a country, it is the **pooling** of these relatively small cushions of wealth into large sums which finances industrial expansion, i.e. provides large capital sums for investment, which in turn should enable the growth of GDP and full employment in a modern economy. . Saving by one sector of the population also enables other sectors of the population to borrow against their future expected income for purposes such as financing a house purchase – another key role of the finance industry.

The cohort of the population that tends to save most consists of those from age 40 and above up to the point of retirement. The group that tends to borrow most on credit cards, loans for purchasing a motor-car and a mortgage for a house purchase is the younger age group. Thus the inter-temporal transfer process is also an inter-generational one.

AN INFLUENTIAL POLITICIAN'S VIEW OF FINANCE

The following may not be a definition of finance but it is a clear indication of what one very senior politician in Europe, Germany's finance minister Wolfgang Schäuble, thinks the function of the finance industry should be and also what it should not be. It probably reflects the thoughts of many other politicians as well as many voters. As the banking industry can only exist within the framework of a 'social contract' with society, what he says is likely to become a key battleground between the finance industry and democratically elected governments, not only in the eurozone but also in the US, the UK and many other countries:

> I believe it is in the interest of the financial sector itself that it should concentrate more on its ***proper role of financing the real economy*** and ensuring that ***capital is allocated in the most intelligent way***, instead of banks conducting the bulk of their trading on their own account.[1]

What is very clear from this statement is the potential conflict that society and politicians see between the role of the financial sector in **facilitating economic growth and full employment through the provision of finance** and the **trading activities** undertaken within this sector. This conflict, which is a fundamental issue covered in this text, has risen to the top of the political/regulatory agenda and has led, amongst other things, to proposals from the European Union for a tax on trading activities. This would be designed to reduce trading activity as well as to raise substantial tax revenue.

[1] *Financial Times*, 31 October 2011. This statement was issued just prior to the G20 summit in November 2011 to be held in Cannes, France (author's italics and emboldenment).

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A VIEW FROM ANCIENT EGYPT

At the level of **saving by a country rather than by an individual**, a good example comes from ancient Egypt. Pharaoh (the king of Egypt) asked Joseph to interpret a dream about seven fat and seven thin cattle, and seven fat and seven thin ears of grain. Joseph interpreted the dream as a warning to make preparations for seven years of hardship and to act pre-emptively during the seven years of plenty by filling the granaries (saving) so that people would not starve when crops failed and famine hit. He therefore advised Pharaoh to store surplus grain during the years of abundance. As economies may also follow seven-year business cycles, such advice would also be relevant to modern times. It is in fact the basis of what is known as counter-cyclical fiscal policy and also macroprudential financial regulatory policy.

Had Pharaoh not taken Joseph's advice, and thus had the country not been able to feed itself, Pharaoh would have had to visit a neighbouring country and ask its leader if he could help by lending grain until the harvest improved. In return, he would have promised to repay more grain than had been 'borrowed'. Today, of course, if a country has not been prudent in its spending and has overspent its income, the same appeal to other countries is made but via the international capital markets. Thus, for example, the US has borrowed massively from China through the first decade of this millennium to make up for its own negligible savings rate. Within the eurozone, Germany, which has high savings relative to its investment in productive capacity, lent substantial sums to other eurozone countries, such as Greece, Spain and Italy, by way of its banks providing loans to customers in these other countries where, on balance, the population spends more than it earns and saves less than is required to finance industry. At some point lenders may take fright when they believe that a country to which they have lent will not be able to service its debts. At this point, we have crises like those from which the eurozone has been suffering since 2010.

FINANCE AS A SCIENTIFIC PROFESSIONAL TOOL

As we are studying **finance as a professional discipline**, we also need a more scientific definition than those given above. A possible definition is:

- Finance is the science of money. It is the scientific discipline that enables us to study, in a rational way, **how to allocate scarce resources** in society. This allocation problem – how best to use labour, capital, raw materials, intellectual property rights and other scarce resources – is key to ensuring the highest level of output, the highest rate of economic growth and the highest levels of employment.
- The finance discipline covers the financing of companies (corporate finance), the financing of government expenditure (public finance), the provision of mortgage finance, pension provision, etc. (household finance), and the financing of payments imbalances between countries (international finance) through the intermediation of the credit and capital markets. While the mechanical processes of finance involve the transfer of investor cash to companies, governments, households and countries, in the other direction they involve the transfer to

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investors of economic risk after a process of risk assessment and risk pricing in the credit and capital markets.

In a market economy, the allocation of **consumption goods** – food, clothes, iPhones etc. – is achieved through the price mechanism. Everything we buy has a price which rises (and may fall) over time to ration it so that supply always equals demand. The price mechanism also operates in the credit and capital markets to ration capital (savings) to be allocated to:

- **Companies** wishing to buy **investment goods** such as machinery, offices, factories, patents, etc. by investors trying to confirm in advance that the company can pay interest and dividends and repay loans from the surplus between sales and costs;
- **Households** wishing to purchase a home by only lending to those believed to have the ability to **service** their loan, i.e. to pay interest and repay the loan;
- **Governments** wishing undertake higher current and capital expenditures than they are willing to try to raise from citizens through taxation.

The interest rate is the most important rationing mechanism in an economy and operates through the PRICE of credit. The starting point for the price of credit is that set by the central bank of a country, which is a short-term rate and which may be called *base rate (UK), repo rate(EU) or, in the case of the US, the Fed Funds rate*. From this rate, **all other interest rates are determined by market forces not by the central bank or the government**. The banker, financier or investor has to estimate, and then price, uncertain (risky) future expected cash flows. The rate which he or she determines as appropriate for this purpose will vary as the central bank rate varies. The central bank rate is, therefore, a key determinant of the demand for credit, hence the amount of real investment in the economy and in consequence the growth rate of the economy. Changes in the central bank rate are also the main instrument used by the central bank to try to manage the inflation rate through the impact of interest rates on levels of demand and on the exchange rate. From our point of view, the key importance of interest rates is that these are what determine asset prices – and **asset prices are what finance is mainly about**. Asset prices include those of securities trading in equity markets and bond markets and of real assets such as residential property.

THE MARKET PRICE AS THE MEANS OF ALLOCATING RESOURCES IN SOCIETY

In finance, **price is the key to almost everything**. For this reason, in a book on investment banking, bank credit, shadow banking and capital markets, we need to examine the **price mechanism** and how it operates through markets – not just financial markets, but markets in a wider sense.

The price mechanism operates in three different types of market, all of which are related:

- Markets in goods and services;

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- Labour markets;
- Financial markets (credit, securities and derivative contracts).

Adam Smith[2] was the first to consider the problem of how the production of goods and services is coordinated so that the 'correct' amount of each is produced (markets in goods and services) and that all of those who want to work can obtain employment (labour markets). Smith believed that the market system ensured that prices moved towards their 'natural' level and would thereby direct human and physical resources to where they were most needed. There was an automatic correction mechanism. If a shortage developed, the price of finished goods would rise and supply would expand as producers attracted more resources (labour and materials) to the production of their particular goods by raising the price offered for these resources. Conversely, if demand fell, the producer would reduce the wages he offered and bid less for (physical) resources and thus both labour and resources would move out of producing his particular goods and move to the production of those where final-market prices were higher. Such a system would be self-equilibrating and resources would thus always be employed fully because prices adjust to changes in supply and demand for resources.

You may notice that no reference is made in this paragraph to financial markets. This is because many people, economists and also the man in the street, have assumed that the financial system and the markets of which it is comprised, are merely a 'veil' covering the 'real' economy. They take the view that a country's output of goods and services is determined by its state of technical knowledge and its access to natural resources. It is somewhat surprising that this has been such a common view since it is also widely acknowledged that economic development is a process of capital accumulation.

Long-term capital enables:

capital widening which is financing to enable the employment of a growing labour force and

capital deepening which involves additional capital being available to each employee, thereby increasing the capital available to each employee and hence (hopefully) productivity.

In addition, the provision of **venture capital** is critical in ensuring that new ideas can be developed into saleable products or services. Thus economic development depends on financial markets – credit and capital – to allocate investment capital amongst competing projects.

In the first decade of the new millennium, there has been considerable evidence that the credit and capital markets have not, in practice, allocated capital in the most appropriate way. In addition, the cost of running the 'capital allocation machinery',

[2] Adam Smith was the world's first economist. He was a great thinker and published the first analysis of the functioning of the market economy when he was a professor of philosophy at the University of Glasgow. His most famous book, *An Inquiry into the Nature and Causes of the Wealth of Nations*, was published in 1776, the same year as the American Declaration of Independence.

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i.e. the cost of the whole financial system, would seem to have increased substantially over this period. The Turner Review,[3] for example, gives a figure for value added (percentage of GDP) of monetary and other financial institutions of just 3.5% in the year 2000 but increasing to almost 6% by 2007. That suggests that the industry became less efficient in its role of intermediation even if we allow for some of the increase involving exported financial services or increased demand for financial intermediation by households. It also reflects, of course, the massive increase in remuneration (including bonuses) over this period paid out to many of those employed in this industry. These issues of 'appropriate' capital allocation and the cost of the mechanisms which enable this allocation are discussed further in Chapter 15 on Market Failure and in Chapter 28 on the Future Structure of the Industry.

A number of economists since the time of Adam Smith have developed what is known as **general equilibrium theory** which posits that competitive markets are efficient and ensure that companies produce the things that people want in the right quantities and at the lowest cost. General equilibrium theory would also posit that any unemployment in an economy must be voluntary since labour is sold in a market and if there is an excess supply this can be overcome by people simply offering their labour for a lower price. We will not enter into discussion of this proposition except to note that throughout Europe and the US in 2012, unemployment is much higher than in many past periods and those who are unemployed would be very unlikely to claim that it was voluntary! We call this market failure – resources of labour which are available are not being utilised to maximise output of goods and services.

When we 'lift the veil' covering the 'real' markets we can look at the issues raised by possible mispricing in the credit and capital markets. When we analyse such pricing what we are in fact pricing is two things:

- The willingness of people to delay gratification today in return for possibly greater gratification in the future (the so-called **time value of money**).
- The risk involved in delaying gratification which is a function of the risk of the underlying investments that people are undertaking and which is measured as the **risk premium** in excess of the time value of money. One risk investors take is that, by deferring consumption, they will have less to consume in the future rather than more (as a result of their investments losing value).

Friedrich Hayek, an Austrian economist wrote in his book *The Use of Knowledge in Society* that 'we must look at the price system as … a **mechanism for communicating information** if we want to understand its real function'. In making their investment decisions, individuals generally have in their mind some model of the financial asset **price determination process** – for example a model of what determines share market prices. The hypothesis to which many market participants and many market regulators adhere – one might call it their mindset – is known as the **efficient market hypothesis (EMH)**. This asserts that financial markets are, in general, **informationally efficient**. This means that prices change instantaneously

[3] The Turner Review was a regulatory response to the global banking crisis, published in March 2009 by the FSA.

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to reflect new and relevant information. This is an example of Hayek's view of the price mechanism – that it transmits information.

We call this process of price adjusting to new information the **impounding of information**; and if the EMH is correct then not only does the price change to reflect the impact of the new information on the asset's future expected cash flows and risk but it happens almost instantly. In consequence, an investor cannot, on a consistent basis, achieve returns in excess of the market return for a financial asset with this level of risk. Another way of putting this is that financial assets are always correctly priced or, to use a common expression in finance, **there is no free lunch**. This means that there is nothing for nothing in this world – no possibility of expected returns above those appropriate for the level of risk. Despite this belief, however, it can be observed that investors still commit substantial resources to trying to 'beat the market', which suggests that they do not believe the hypothesis to be true, at least for them. Logically, analysts in the City of London and so-called active investors must believe that the EMH is false.

Above, we used the expression '**always correctly priced**'. However, this can mean two things:

- It may mean that every financial asset is correctly priced **relative** to other financial assets, such as Google shares relative to Intel shares;
- It may also mean correctly priced in relation to **fundamentals**, which means expected future cash flows from the asset and the risk of them not materialising.

In Chapter 15 on **market failure** we will consider in some detail whether credit and capital markets do in fact seem to misprice financial asset classes in terms of fundamentals, leading to the mispricing of whole asset classes such as housing, the stock market, structured assets and peripheral eurozone debt. Such mispricings, when market prices rise beyond what a scientific valuation based on fundamentals would suggest to be appropriate, are known as **bubbles**. Under the assumptions of the EMH such mispricings should not happen.

The reason why it is important that we know if markets are informationally efficient in **both** the above senses is that if markets always price asset classes correctly, for example equity shares as a whole (not just one equity relative to another) based on fundamentals, then we may assume that the resulting set of prices will also result in **allocative efficiency**. This type of efficiency means that the credit and capital markets will allocate capital (savings) to their most productive use and thus, other things being equal, the output of the economy and household utility will be maximised. Allocative efficiency implies that the price of an asset accurately reflects the (appropriately risk-adjusted) discounted stream of future earnings that it is expected to yield over its lifetime.[4] Thus even if markets are informationally efficient, if they fail to satisfy the allocative efficiency condition, then resource allocation decisions (such as residential construction prior to the house price collapse in the US or spending by some peripheral eurozone governments prior to 2010)) can result in significant economic inefficiencies. There is evidence that the risk adjustment to the discount rate that has been undertaken on

[4] I cover the concept of discounting and risk adjusted discount rates, below.

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the pricing of a number of asset classes in, for example, US housing, US mortgage debt and peripheral eurozone sovereign debt, has not been appropriate for the risks that could have been foreseen. These risks were, in all cases, foreseen by a not-insignificant number of key analysts in central banks and research institutes. However, their views were not incorporated into market pricing. When times seem good, nobody wants to hear bad news!

NON-MARKET ALLOCATION MECHANISMS

An alternative means of rationing goods and services and deciding on the allocation of capital is through bureaucratic decisions such as those that were made by state bureaucrats in societies operating under communist principles. In such societies, Karl Marx said that the rule for allocation should be '**From each according to his ability, to each according to his needs**'. This clearly requires a bureaucratic system to assess such needs and allocate goods and services (motor car and iPhone allocation, for example) on the basis of this assessment. It also then requires the necessary investment in industry, yet without a mechanism to suggest what goods and services industry should be producing. In addition to questions about the efficacy of such an allocation process there also arose the difficulty of motivating people to be more productive if the financial benefits of working to the maximum of one's ability did not, in large part, flow to the person putting in the effort. Thus fairness was by no means guaranteed. In states with such allocation mechanisms there is little incentive to innovate or work hard.

Until the late 1950s, China had a system of resource allocation based on central planning. But in 1957, Hsueh Mu-chiao, who was then chairman of the State Statistical Bureau, advocated that central economic agencies should retain planning responsibility for only a few areas, mainly macroeconomic, and that other decision-making on resource allocation should be decentralised. Apart from a few key products, production planning for all other products, he said, should be determined by individual enterprises according to supply and demand. Decentralisation of decisions on resource allocation and rationing by price, based on supply and demand, is a key feature of capitalist economies. In China this early step towards a greater use of the **pricing mechanism** as distinct from central bureaucratic fiat was an important one in starting the process which transformed the country's' economy. However, it was not until after Chairman Mao's death, when Deng Xiaoping took over the leadership of the country, that economic reform moved ahead rapidly. His leadership led to ongoing reforms which have resulted in a dramatic increase in China's GDP

In China, until quite recently, housing was allocated by the state, not through the market system. In an analysis of the Chinese property market[5] the author notes:

> Before 1998, China did not have a residential real estate market to speak of and there was no need for such exotic financial products (mortgages). In urban areas all housing was built and allocated by the state through the ubiquitous 'work unit'.

[5] Jamil Anderlini, 'A Lofty Ceiling Reached', *Financial Times*, 14 December 2011.

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Even today, however, the People's Republic of China is not a pure capitalist economy. There remains much direction of credit and capital from the centre. Even the major banks in China, while they have quotations on the stock market, remain subject to a considerable degree of direction from the state as it remains a majority shareholder.

OPTIMAL ALLOCATION OF RESOURCES OVER TIME IN AN UNCERTAIN ENVIRONMENT

The word 'resources' in the expression '**optimal allocation of resources**' generally means real resources – labour, raw materials, finished goods and services, etc. But prior to this stage in the resource allocation process, the finance field is concerned with the optimal allocation of savings, i.e. cash, from those who save to those who can make a good case that they can use the savings in an 'appropriate' way. The finance process is about ensuring that only those who can make the 'best' economic use of the real resources that cash can buy are allocated savings belonging to others to enable them to acquire such resources. A company, for example, has to be able to demonstrate that it will make good use of the financial resources provided by investors; and thus one of the functions of the 'scientific' financial system is to provide a means of deciding what is a 'good' or 'appropriate' use.

A good example of effective resource allocation would be the start-up of the Apple Corporation by Steve Jobs. He was able to create what is now a large and successful company by persuading others to provide the cash necessary to expand it from its origins in the 1970s as nothing more than a 'clever idea' he developed at home to becoming the world's most valuable company. The individuals who provided the initial financing must have believed that parting with their cash by investing in Apple could generate an acceptable return to reward them for giving up spending power now (time value of money) **and to compensate them for the risk** they were taking. The first 50 Apple 1s were built in Job's parents' spare bedroom. Just a few months later Apple moved upscale – to Job's parents' garage!

There were two types of resource allocation in the process of Jobs scaling up his 'brainy idea' into something that could become a real company. One involved investors allocating their scarce savings to provide the cash needed to buy shares issued by Apple. The other was the use of the cash realised from the sale of shares for the purchase of real assets and the payment of wages by the company. However, these two types of allocation must be linked. Investors will only allocate their savings if they know what these resources will be used for and if they believe that they will generate a satisfactory return, given the risks involved. Thus there are two stages in the investment process.

Financial investment

In the first stage, households allocate their spare financial resources (savings) to **financial** assets such as buying shares in Job's start-up venture. A financial asset is simply a **contract** in the form of a document – the **prospectus** – issued at the

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time of a new share issue (an initial public offering of shares generally known as an IPO) or at the time of a bond issue (a debt contract). The contract agrees the transfer of household cash directly to companies in exchange for shares or bonds at prices based on the future cash flows that the real assets are expected to generate and the risk of these case flows not materialising.

However, households will only transfer resources through share or bond purchases if the information provided in a prospectus convinces them that the company will make 'good' use of the resources. Thus the company has to lay out a business plan detailing the use to which the proceeds of the financing will be put, i.e. the real assets to be purchased, the output expected from the use of these real assets, the number of units expected to be sold and their expected price. From this it is possible to estimate **future cash flows**. There then needs to be a **scientific means of evaluating** whether this is a 'good' use of resources which will also involve trying to assess the degree of certainty that we can have that these cash flows will actually materialise. I cover methods of analysis under 'the three pillars', below.

Real investment

In the second stage, companies allocate the cash raised from investors to purchasing scarce **real** resources, i.e. investment goods such as factories, power stations, machinery, research and development, as well as hiring labour on the basis of the business plan which they put to investors. As decisions on real investment are simply the counterpart to decisions on financial investment, methods of analysis are also covered under 'the three pillars', below.

TIME SHIFTING OF CONSUMPTION

While finance is widely taught as being principally about how to finance new investment in industry (as in our Apple example above) it is also, and very importantly, about **changing a household's consumption pattern relative to its pattern of income earning**. Optimal resource allocation over time for a household may involve the time shifting of consumption, i.e. changing an individual or household's allocation of consumption resources over a lifetime by borrowing and investing. For households, therefore, **optimal resource allocation over time** means **maximising household utility** over time, i.e. achieving the greatest satisfaction from income over a lifetime through borrowing and lending/investing. This may mean, as we noted earlier, borrowing when young and when income is low in order to have more to spend on holidays, a motor car, housing, etc. and paying the price of consequently having a lower proportion of a (higher) income to spend in mid-life due to interest payments and capital repayments. It also means increasing saving in mid-life in order to have an income in retirement from investments.

RISK BEARING

In addition to considerations of consumption, different people have different attitudes to risk bearing and therefore to the question of how much of society's risk

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they are willing to absorb by, for example, buying risky shares. Thus, in addition to enabling an optimal allocation of financial resources (cash), finance is also about the optimal distribution of **society's economic risks** away from those who do not want to bear them to those who have some comparative advantage in bearing them in exchange for an appropriate expected rate of return to compensate for risk bearing. Undertaking the pricing of risk, i.e. determining the price at which risky shares, bonds, companies, etc. should be bought and sold, is a key activity of people working in the finance industry. Those who buy shares or bonds are taking on society's economic risks so others do not need to bear them. The major economic risks manifest themselves as credit risk, equity risk, currency risk, interest rate risk and commodity price risk.

A problem to which I allude in the text is the increasing unwillingness of Western financial institutions and individuals to bear all the equity risk necessary to finance the hoped-for rate of economic growth in the West. In consequence, the equity risk of Western economies is being increasingly borne by financial institutions and individuals in Asian and OPEC countries.

THE THREE PILLARS: THE SCIENTIFIC TOOLS OF FINANCE[6]

Finance has become a scientific discipline since academic finance research began in the United States in the 1950s. This has resulted in finance changing from being a field where there was no 'scientific' way of making evaluations to one where scientific evaluation is central to the whole business. There are three essential elements to this science. The three pillars are: first, a means to **optimise over time**, i.e. to analyse the trade-off between having money now and having more in the future; second, a means of **measuring and managing risk**; and third, based on these first two, a means of **valuing financial assets**, i.e. pricing shares and bonds, which has the role of effecting the allocation of scarce resources in society and effecting the trade-off between having money to spend now or having more to spend in the future.

Pillar 1: optimisation over time; the time value of money; discounting and compounding

The key financial decision that has to be made by decision-makers in firms is whether spending money now on something which will only bring benefits in the future is justified by the expected future benefits. If they are spending money on, for example, an expansion in productive capacity, their hope is that they can sell the additional output in the future at a price which will reward appropriately those who financed the company.

The same type of decision-making process is also used by individuals, as in the example below. But for individuals, as well as having to make decisions on investment (buying a house or investing in education), they also have the possibility of **consumption shifting decisions** – borrowing to consume now in exchange for paying interest and repaying a loan in the future and therefore having less

[6] This is the approach taken by Merton and Bodie in *Finance* (Prentice Hall, 1998).

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consumption in the future. This is something companies do not do as they do not consume. Governments also have this option of bringing consumption spending forward. Government borrowing represents a decision to finance expenditure, perhaps on unemployment benefits or health care, not from taxes raised from today's taxpayers but from future taxpayers. It is thus what is known as **intergenerational transfer**.

Example: an individual's investment in his or her own human capital

An example of an investment decision that an individual has to make is one where someone who is 21 years old and has completed a bachelor degree is considering whether or not to take a second degree, perhaps an MSc in Finance. The course fee is, say, £18,000. The additional cost of accommodation and food for a year compared with living with parents is, say, £6,000. As an alternative to undertaking the MSc, the person could choose to take a job immediately after his or her first degree with a salary of £21,000. Thus the **cash cost** of undertaking the degree programme for one year is £24,000 plus an **opportunity cost**, i.e. income foregone, of £21,000, giving a total investment in the project (obtaining an MSc) of £45,000. If money is borrowed to finance the programme, then there is also an explicit interest cost to be considered. If the money does not need to be borrowed and is already available, then the alternative would have been to invest it in a bank deposit or in the stock market. There is thus an **opportunity cost**, i.e. a potential return that is lost, to be added here also.

To make that investment worthwhile, the investor has to estimate the increase in future earnings that this investment in **human capital** will bring and over what period those additional earnings will accrue. If, for example, the higher earnings did not accrue until the person was very senior, perhaps not until he or she was over 50 years old, this would be very different from a situation where the higher income commenced immediately after graduation – **timing of cash flows is very important**.

The decision on whether or not to undertake the degree and other decisions of this type involving future cash flows and time can be aided by taking account of the **time value of money**. This simply means that money today is more valuable than the same amount at a future date. Mainly this is because money can earn interest or, looking at it the other way, we can obtain money now (spending power) if we are willing to pay interest on that money in the future. In effect, a loan is a way of buying **liquidity** now, i.e. spending power for which the price we pay is the interest cost for each year the loan is outstanding and the loss of spending power sometime in the future when we have to repay the loan. But equally the time value of money enters the equation since most people would prefer to benefit from a higher income immediately, rather than have to wait until they are 50 years old for the higher salary to be paid.

Example: corporate finance, a capital budgeting decision by managers

Companies have to make decisions on what projects to invest in. A mobile phone company, for example, might have more demand for existing phones than its

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maximum output but may also want to develop new phones, for example a smartphone. It thus has *two* investment opportunities – to lay out money now for a capacity expansion or alternatively for a new product (or, in an ideal world, both). The **capital budgeting department** has to estimate the investment cost involved in each and the likely revenues to be generated each year until the project comes to an end. On the basis of these projections of future cash flows they can then discount them to a present value. The discount rate that will be used is the firm's **cost of capital** which is normally taken to be a weighted average of the cost of equity and debt and which reflects how risky the firm is. A good example of such investment decisions can be seen in the case of Nokia, once the world's largest mobile (cell) phone maker. It invested heavily in developing a 'smartphone' to compete with the Apple iPhone. Despite a very large investment, it did not succeed and, believing that the project would not generate the required return for its investors, abandoned it in 2011 in favour of buying in technology from Microsoft.

Another case study of a different manufacturer of mobile phones is LG Electronics, the world's third-largest handset maker by sales, which planned to raise about 1,000 billion won ($892 million) in a rights issue to help turn round its troubled smartphone business:

> 'The stock sale is aimed at securing financing for investment in our core businesses. We plan to revive our global competitiveness in key areas such as smartphones through steady investment,' the company said. Although its cross-town rival Samsung Electronics has caught up fast in the high-margin smartphone segment to become the world's largest smartphone manufacturer, LG remains a marginal player in the competitive market. LG did well with so-called feature phones before Apple shook the market with the iPhone. But it has fallen behind in recent years as it misread the mobile market and is still suffering from its late arrival. 'Investors are concerned more about its fundamental competitiveness, rather than its short-term cash flow,' said Choi Nam-gon, analyst at Tongyang Investment Bank. 'And the stock sale plan has sparked fears that its handset business may be in worse shape than they thought.' The announcement sent shares in the company down 13.7%.[7]

Clearly the above case suggests that investors (one side of the savings/investment symbiosis) do not have confidence in management's resource allocation decision (the other side of the symbiosis). Allocating a huge amount of financial resource to revive LG's global competitiveness in a highly competitive business sector must be thought by investors to be inappropriate given the share price fall on its announcement. 'Inappropriate' means unlikely to generate investors' required rate of return and hence be likely to reduce what is known as **shareholder value**. Capital market pricing should prevent such misallocation of real resources. In this case the company seems to be going ahead even though investors clearly do not think it is in their interest as the owners of the company.

[7] Song Jung-a, 'LG Eyes Rights Issue for Smartphone Turnaround', *Financial Times*, 3 November 2011.

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Capital budgeting

Investment decision-making in a company, such as investing in developing a new phone to try to compete with Apple, should be undertaken using a **capital budgeting** process. This involves calculating the **net present value (NPV)** of a project. This is an amount calculated by deducting the present value[8] (which means allowing for the timing of cash flows) of the real investment (outflow) from the present value of all future cash (inflows) generated by the project. To **maximise shareholder value or wealth**, the investment decision-making criterion would be to accept all projects which generate a **positive net present value**, which means that the project would increase **shareholder value** (note that the word 'net' means the present value of future cash inflows minus the present value of cash outflows). This means that such projects should increase the value of the company's shares. If two projects have a positive NPV but the company does not have sufficient internal resource (retained profit) to finance both, then, other things being equal, it should raise additional funds on the primary capital market or from a bank.

Making corporate finance decisions, such as whether to invest in the design and manufacture of a new type of cell phone or financial investment decision on how much to put aside to provide a secure income in retirement, involves discounting (bringing to a present value) and compounding (to a future value), respectively. An obvious question, however, is what rate of discount should be used. In essence this is asking, first, what is the **pure time value of money** (i.e. the cost of money for a particular period of time if there were no risk at all of it not being repaid) and, second, what is the **risk premium** (i.e. the extra return in excess of the **pure time value of money**, necessary to persuade investors to take on any particular risk). The pure time value of money is generally taken to be the yield on government bonds for the period in question. The reason for this is that government bonds, at least those of developed countries, have in the past been assumed to be risk free. We now know of course, since the eurozone crisis, that this is no longer true. For the eurozone, we would instead take the yield on German government bonds (bunds) as the risk free rate of interest. We then need to add to this the additional amount of return that investors require to compensate them for the risk of the particular financial assets they are considering buying, which we call the risk spread.

Pricing risk to assess the appropriate risk spread is the key function in finance, and from that comes the ability to price shares, bonds, loans, insurance policies, options, corporate acquisitions, etc. In the financial intermediation industry, pricing is everything – indeed it is what most of the industry is about. Get the price right (for the risk involved) and you make money. Get the price wrong and you lose your job! If prices are very wrong, they may cause global financial instability such as we have seen since 2007.

Pillar 2: risk management

Risk is the **possibility of unexpected outcomes**. In finance, risk is generally measured as variability of return, i.e. the likelihood that the **actual return** on an

[8] See a textbook for more on present values, discounting and capital budgeting.

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asset will be *either* higher or lower than the **expected return**. The expected return will generally be estimated by taking historical returns on the asset as a starting point. Uncertainty as to the actual return that will be generated is known as **variance**. The wider the range around the expected value of cash flows or prices that are expected, the riskier is a project, an investment or a portfolio.

How do different people view risk?

Return is good; risk is bad in the sense that it raises the possibility of doing much worse than expected and indeed of losing everything. But generally we expect that the higher the estimate of risk on an investment, the higher will be its expected return. Risk management can be defined as a method of identifying risks, measuring and assessing them, formulating the cost–benefit trade-off from risk reduction and deciding on what actions to take. These actions in turn depend on individual attitude to risk, which is known as the degree of **risk aversion** of an individual or a company. This is a measure of how willing someone is to pay to reduce their exposure to risk (for example by taking out insurance versus remaining uninsured).

For risk-averse individuals and households, some form of risk management may raise their utility by reducing the uncertainty of future income and hence consumption. Thus risk management techniques and products available to households, including insurance, can increase overall welfare.

Given that investing now for expected future returns is risky, the second pillar of finance is a set of methodologies for controlling the risk to which an individual, household, firm, government or financial institution is subject as a result of holding financial assets. This may involve, for example, **portfolio diversification** as employed by asset managers as well as self-directed investors, **insurance** against risk involving a payment to someone else to take on a particular risk, or the **hedging** of the risk by offsetting it in some way. The financial services industry can help individuals and companies with assessing and managing risk.

While financing is often thought of as the transfer of cash between investors and firms, in fact it can equally and often more usefully be thought of as the transfer of risk. Investors want to take on the various risks that exist in society because **society rewards risk taking** and they wish to participate in that risk/reward system. But as risk means the possibility of loss, risk management to minimise the possibility of **catastrophic loss (such as could result from investing all one's wealth in a single company which goes into liquidation)** is important for most people and also for most financial firms.

Pillar 3: asset pricing

The key function in finance is pricing assets and, most importantly, pricing them correctly for the level of risk. For an intermediary, '**getting the price right**' is the key to success. Correct pricing is necessary in the provision of bank loans (the price in this case being the interest rate charged for the level of risk on a particular loan), in initial public offerings of shares and bonds (the offer price), in secondary market transactions in shares and bonds (bid and offer prices), and in making acquisitions in the merger and acquisition market (the acquisition bid price).

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Cash market instruments (capital assets)

These are assets, the value of which derives from the cash flows they generate – either interest or dividend payments. These cash flows can be valued using discounted cash flow techniques. The main capital assets are equities, bonds and loans. Equities and bonds are traded in **cash** markets (so-called to distinguish them from derivative markets). Loans are also traded, particularly amongst banks and institutional investors, but there is no public market as there is for equities and bonds. Cash market instruments are also known as **capital assets**, which always have value provided they are generating cash or can be sold to realise cash.

A government bond is a liability issued by a government (an asset to the holder, i.e. the investor) on which the government promises to pay a fixed amount of interest each year for an agreed number of years and at the end of the agreed period to repay the original amount (the capital or **principal**). Such a bond which provides regular income is known as a **coupon** bond, a term which derives from bonds originally being paper certificates with small coupons attached (rather like small banknotes) which could be clipped off the bond and taken to a bank (a paying agent) for payment in cash.[9] When an investor asks what the coupon is on a bond, he or she is asking what is the original rate of interest promised on the bond. The original investor in the bond receives the annual coupon in return for paying the **issue price** or **par value** of the bond. It has conventionally been assumed that the cash flows arising from a government bond are free of risk since governments of developed market countries (sovereigns) are conventionally assumed to be extremely unlikely to default on payment though today that assumption may not be valid.

A corporate bond is a liability issued by a company (an asset to the holder) on which the company promises to pay a fixed amount of interest each year for an agreed number of years and at the end of the agreed period to repay the principal. In this case, however, there must be some degree of risk involved as we cannot be sure that the company will in fact meet all its financial obligations. This is known as **credit** risk. In consequence, investors expect to receive an amount of interest in excess of that on an otherwise comparable government bond. The additional amount received is known as the credit risk premium. A key issue in valuation is how to assess the risk premium required on any liability being issued in order to make it attractive to investors.

Financial assets, i.e. liabilities issued by governments or companies, have value to investors only because they can or actually do generate income (cash flow). These cash flows are known as **P&I** flows which can stand for either **principal and income** or **principal and interest**, where 'principal' means the repayment of part, or the whole, of the original investment, and 'income' means interest and dividends. There are three sources of P&I available which can service loans, bonds and shares. However, if we include real estate assets, which are also a very important class of asset, then there is a fourth (which is listed last in what follows). They are:

- **Income from employment** by individuals (to service mortgages, credit card debt, etc.). This is income generation from **human capital**. We assume that if

[9] Karl Marx in *Das Kapital* referred disparagingly to 'mere coupon clippers', i.e. financiers, in contrast to genuine entrepreneurs.

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someone invests in, say, a Master's degree in finance, his or her human capital is increased and thus the expected future income flows from employment rise.

- **Cash flow from the operations** of a company or business (to service loans, bonds and equity).
- **Tax revenue** raised by the finance ministry of a country (to service government debt).
- **Rental income** from property (to service the mortgage on the property or to provide income to the owner).

Derivative market instruments (contingent claims)

This is another type of financial instrument which is closely related to capital assets. These are instruments which derive their value from the value of the **underlying** assets on which they are based, which may be a capital asset or a real asset such as oil or wheat. They are a means of exchanging risks and are thus not assets in the traditional sense. They are not capital assets as they do not generate cash flows. At their **expiry** they generate either a profit or a loss depending on the change in the value of the underlying asset. They are often known as **contingent assets** as their value is based on developments in the future, i.e. future economic conditions and the impact this has on the underlying asset (capital or real). They are a way of 'placing a bet' on an outcome with, for example, two people with opposite views taking opposite sides of a derivative contract. There is no generation of principal or interest or dividends. There is a simply a transfer of cash between two traders when a contract expires.

Real assets

Finally, there are **real assets**. This category includes real estate which we covered above. But it also includes commodities. There are two main categories of these. Soft commodities are mainly foodstuffs, and hard commodities includes metals and plastics (e.g. copper, aluminium, steel, steel bar, oil, polypropylene, polyethylene) used as inputs to the manufacturing, transport and building industries. There is also oil which is a major input to manufacturing, transport and home heating. Commodities are traded on specialised commodities exchanges such as the Shanghai Futures Exchange and the London Metal Exchange. But in addition there are assets which are principally just **a store of value**, such as works of art, gold, silver and precious stones. Such assets have **no cash flows associated with them** and have only very limited value as inputs to other processes. Most gold is held as a store of value which may be in the form of jewellery and not for utilitarian purposes such as the plating of electrical contacts. The value of such assets is derived from subjective visual enjoyment or belief in the value of the asset as a store of value in times of uncertainty. Their value cannot be calculated in the way we scientifically calculate the value of a capital asset from its expected cash flows and risk. Such assets are, therefore, **purely speculative assets**, unlike company securities which generate cash flows.

In finance a key function is estimating the value of an asset. This is a complex matter and one to which whole modules in finance degrees are devoted. For those

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who have not covered these issues elsewhere I would recommend referring to a good textbook on valuation.[10]

THE UNIVERSE OF FINANCIAL ASSETS IN WHICH WE ARE INTERESTED

Figure 1.1 shows the global stock of financial assets held by investors, i.e. liabilities issued by companies, banks and governments and, indirectly, by households as of year-end 2006, just before the financial crisis hit, resulting in a great reduction in these asset values. This is the global set of assets in credit and capital markets which is the focus of this book.

ASSET VALUATION MODELS

The Law of One Price, arbitrage and the price of financial assets

The **Law of One Price** is a statement that, in a competitive market, if two assets are essentially the same they will trade at the same market price. This result comes about because of the ability of **economically rational** traders to conduct **arbitrage trades**, meaning the purchase of one asset and the immediate sale of another equivalent asset in order to generate a riskless profit from the price difference between them. This is a form of **pairs trading** where two transactions, involving a sale and a purchase, are undertaken at the same time. For example, if the price of foreign exchange (**forex**) in London is different from that in New York then, since there are no transport costs involved, it is very easy and cheap (apart from commission payments) to buy in the cheaper market and immediately resell in

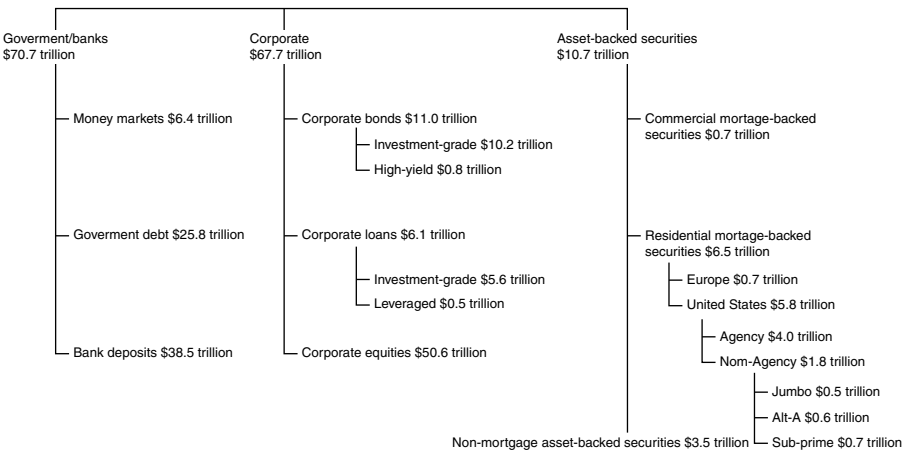


Figure 1.1 Global financial assets as of year-end 2006

Sources: Bank of England Financial Stability Review (FSR), October 2007 (p. 20), http://www.bankofengland.co.uk/publications/Documents/fsr/2007/fsrfull0710.pdf (accessed, 2 February 2012). BIS, Board of Governors of the Federal Reserve, European Securitisation Forum, Eurostat, Fitch Ratings Ltd, McKinsey Global Institute, ONS, Securities Industry and Financial Markets Association, Standard and Poor's, World Federation of Exchanges and Bank calculations.

[10] For example, Zvi Bodie Alex Kane, Alan J. Marcus, *Essentials of Investments* (McGraw Hill/Irwin, 7th edition, 2008).

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the higher priced market. We can use the law of one price to make the assumption that **arbitrageurs will ensure that this 'law' holds in relation to the values of assets that are not just identical but are similar to each other** and which can be easily and cheaply traded. Of course arbitrageurs may not have sufficient capital available to undertake the required volume of pairs trades to ensure that this 'law' always holds. If so, this would be likely to result in the law of one price not in fact always holding in practice and consequently to the mis-pricing of assets.

The Law of One Price is the most fundamental valuation principle in finance. In the case of bonds, the 'law' would say that if two bonds have exactly the same cash flows and the same degree of risk, even though they may have been issued by different companies, they should trade at the same price (or more correctly, yield, which is the expected return), because if one were trading at a higher yield than the other, investors would buy it (it has a lower price) and sell the one trading at a lower yield (it has a higher price), thus bringing about an equality of yield.

Valuation models

If we were always comparing an asset with others that were identical to the one we were trying to value, we could use the Law of One Price directly and easily. In practice we are unlikely to find an asset which is exactly the same as the one we are trying to value. We must, therefore, employ some quantitative method to estimate value from **the prices we do know** which are of comparable but not identical assets. Quantitative methods that we use to **infer** the value of an asset, based on information about the price of a comparable asset or asset type and market interest rates, are known as **valuation models**.

Valuing real estate

Real estate valuers, when they visit a domestic residence to undertake a valuation, consider information on the value of houses in general in the particular area (is it a high or low-valued part of town?), the floor area, the condition of the house, but most particularly the transaction prices at which sales have been achieved on **comparable** houses in that area. If a house next door which is virtually identical has sold for a particular price, then that would be the starting point. The next step would be to consider the relative condition of the two houses. The step after this would be to take the difference in timing between the two sales and adjust for the general change in house prices in the area over this period. In practice, of course, two houses are seldom very similar (or two office blocks or shopping centres) and thus the process is much more complex. Thus real estate valuers have to undertake an extensive education programme to gain their professional qualification from, for example, the Royal Institute of Chartered Surveyors.

Valuing equity shares – the price/earnings (PE) valuation approach

One approach to the valuation of a company's shares is by comparison with similar companies using what is called the price/earnings multiple approach. The **price/earnings (P/E) ratio** is the market price of the share divided by this year's actual or next year's expected **earnings**, i.e. profit, not dividends. This can be thought

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of as the number of years it would take this company to generate profit per share equal to today's price per share (assuming no growth in profits). If profits are likely to grow fast, this number will be large (high P/E) compared with a company which is not expected to grow fast (low P/E) since, in a growth company, by definition earnings (profits) are expected to grow very rapidly, thus quickly reducing the number of years needed to generate the price of a share from its earnings.

If we are using this approach to value one company which is not yet trading on the market, for example if it is going to make an initial public offering of its shares, we look for other companies which are trading on the market and which are comparable – same industry, growth prospects, etc. On the basis of the law of one price we simply multiply the current (or next year's expected) earnings by this multiple to obtain a price. If the two companies really are comparable, then when the company is floated on the primary market at this price, this will also turn out to be the price at which it continues to trade in the secondary market i.e. the price will be right

If we look in the Companies and Markets section of the *Financial Times* we can find the average price earnings ratios of a wide range of countries and companies. For example, in the UK, the price earnings ratio for the market as a whole at the time of writing was 10.8. This means that an average company would take 10.8 years to earn in profits an amount on one share equal to the value of that share (assuming no growth in earnings). At the time of writing, Peru had a P/E ratio of 38 while Portugal's ratio was only 4.8 – both very different from the UK.

It is possible that these three countries, UK, Peru and Portugal, all have similar companies to each other which are expected to grow at similar rates (though it is not actually very likely). However, the investors in each country may put very different valuations on these similar expected profit streams. This may be because investor **attitudes to risk differ** by country. Of course it is actually more likely that the reason for the different P/E ratios is that Peruvian companies are expected to grow much faster than Portuguese companies, with UK companies somewhere in between.

Firms with a high price earnings multiple are expected to have a high rate of growth of earnings and hence of dividends. This must mean that they have excellent new high return (i.e. higher than their cost of capital) investment opportunities available to them. When the managers of such firms undertake a capital budgeting exercise, they will find many projects that have a positive net present value and hence which should be undertaken. Apple Corporation would be an example of such a company.

Valuing equity shares – book value multiple

If you own a portfolio of shares and revalue them daily, then the value recorded in your company accounts (book value) will be the same as market value, which is the price at which they are currently trading on the stock market. If you don't revalue, book value and market value will diverge. The same logic applies to companies.

Book value is the amount at which a company's assets were originally recorded in its accounts. This is also known as **historical cost**. Over time book value (historical cost) and market value (i.e. the value of the company on the stock market) may get out of line. We would expect that the stock market value would be at least

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equal to, or above, book value (provided book values are correct values), since if it is below it could be profitable to wind-up (liquidate) the company if its assets could actually be sold for book value. The proceeds, which would then exceed the share price, would be returned to shareholders.

We generally expect the share price to be higher than book value since any company is supposed to have some competitive advantage in using its assets to make them worth more in terms of enterprise value than simply as stand-alone assets of various types. Another way of putting this is that companies should have **franchise value**, which is the amount of value their brand name, reputation and skills add to the company value beyond the cost of their assets.

Discounted cash flow valuation of bonds and equities

An asset's fundamental value arises simply from expected future flows of cash back to the investor from the asset. To value these, we have to consider the time value of money and the likelihood that the asset will in fact generate these cash flows (risk). Thus we have to use an appropriately risk-adjusted interest rate to discount these cash flows back to a present value.

A bond is just a contract between a government or a company and an investor. In the case of a fixed interest bond (the most common type) the contract specifies that the borrower will pay the investor the same fixed amount of interest each year of the bond's life and, at the end of its life, the initial investment will be repaid.

Let's start with (supposedly) riskless cash flows from a government bond rather than a company (corporate). Such a bond involves an investment now of perhaps $1000 in return for annual interest payments, known as **coupon payments** of, say, 5% i.e. $50. However, although we paid $1000 for this bond, during its life, its price will change continually as a result of **market** interest rates changing. We have to consider why this happens when the cash flows from the bond have not changed. We might, for example, want to know what our bond is worth 2 years after buying it i.e. 8 years before the government pays us back our $1000 because we need to sell it as we want the cash to go on holiday.

We know the annual interest rate on the bond i.e. that a payment of $50 will be made every year regardless of the price of the bond in the market and we know the **redemption value** i.e. the **face value** of the bond (1000) that will be repaid at maturity i.e. in 10 years' time for a 10 year bond. Thus we know all the cash flows for **certain**. What we don't know is the **rate** at which cash flows should be discounted (the discount rate) to calculate the value of the bond **today**, let's say 2 years after issue.

This rate depends on the time value of money over the remaining life of the bond (8 years) which can vary day by day. We can find this rate in the market. Market interest rates are the current rates that can be observed for comparable bonds and which will provide us with valuations of **comparable assets** to enable us to price our particular bond by using the Law of One Price. Thus, if we look at the prices at which other government bonds with a remaining life of 8 years are trading (comparable assets), we can find the rate at which investors are discounting future cash flows which are **certain** in order to come to the market price. If other government bonds with a remaining life of 8 years (or new bonds being issued with an 8 year life) are giving a yield of 7%, we then know that for our particular

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bond to be attractive to investors, it too must **yield** 7% i.e. give a return over its remaining life of 7%. If it didn't, no one would be willing to buy it from us. Since the **coupon** on our bond is only 5%, i.e. the annual interest payment, we know that we can no longer be sold for its purchase price (1000). However, when it is **redeemed** i.e. paid back by the government, the **redemption value** also known as **face value** or **par value** will still be 1000 i.e. the investor to whom we sell it will be repaid 1000 by the government as this was the original issue price. Thus the new investor will receive $50 of interest for each remaining year of life of the bond plus a capital gain at redemption which, **in combination** in present value terms must give a return of 7% per annum. The price which achieves this is approximately $880.[11] At this price, the bond gives a **current yield** (or **running yield**) of around 5.7% annually but when the future value of the additional repayment amount above the purchase price is factored in at its present value, the **yield to redemption** of the bond is found to be approximately 7%. In theory investors are indifferent between receiving a current yield of only 5.7% followed by a capital gain at maturity of $125 and receiving a running yield of 7% and no capital gain at maturity. This is because once the time value of money is allowed for, both sets of cash flow have the same value to the investor.

Remember: the price of a bond moves inversely to the market rate of interest

As our example above shows, if the interest rate for eight-year money **rises** above 5%, the price of our bond will be **falling** so that anyone now buying that bond will obtain a higher yield (return on amount invested) than we who bought it at its par value as a result of paying less for it yet is still receiving only 5% fixed interest (based on the issue price of 1000). If there is no 8 year bond to which we can look to find the market yield for this term, we can extrapolate from rates on bonds with a final maturity on either side of 8 years.

We can apply the same principle to risky corporate bonds. We can look at the rate at which investors must be discounting expected cash flows on a bond of similar final maturity date and similar risk (a bond with the same rating from a credit rating agency) and then discount the cash flows on the bond that interests us at the same (risky) discount rate. If this bond is not trading in the market this enables us to calculate the price we should be willing to pay for it. It is very difficult to cover bond valuation in a few, condensed) paragraphs such as those above and probably even harder to feel comfortable from reading those paragraphs.

In the case of equities, there are a number of discounted cash flow approaches to valuation such as the dividend growth model. For the sake of brevity we will not cover these here. For a full treatment of valuation of equities and bonds you might wish to refer to a standard finance text such as Bodie, Kane and Marcus *Essentials of Investment* published in 2008 (7th edition) by McGraw Hill.

[11] Calculator available at http://www.montegodata.co.uk/scripting/bondvaluecalculator.htm

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ISSUES IN BANK ASSET VALUATION AND VALUATION OF BANK SHARES

As we are studying financing institutions, banks are a particular type of company in which we have an interest. A key problem with valuing financial assets held on the balance sheet of financial intermediaries and banks in particular is that their value depends not simply on objective facts, i.e. cash flows and risk adjusted discount rates, but also on accounting and regulatory rules. In fact, there are two quite different ways of valuing financial assets, such as loans and securities when these are held on bank balance sheets.

Historic or amortised cost

The first, and more traditional, method is amortised cost – an accounting rule based on the historical cost of acquiring the asset. Using the historical/amortised cost principle, the value of an asset would *not* be increased if its market value rose, since under this principle we use acquisition cost (historic cost) as the basis of valuation. Equally, accountants within banks do not need to reduce the value of the asset from its historic cost, i.e. amortise it, unless there has been an **impairment** to its value. Impairment is something which reduces the value of the asset. It is in trying to define what counts as impairment that the difficulties in the area of valuation have arisen.

The traditional rule has been that for loan assets held on the bank balance sheet and **intended to be held to maturity** there has to be some 'event' that **triggers** a reconsideration of the loan (or other asset's) value – something that suggests that the asset has become impaired. But provided a borrower continues to pay interest, even if we know that he or she will be unable to repay the loan, we do not recognise impairment. Equally, even *if* the loan could to be sold to another bank and we know that on sale it would have a value lower than the historic cost, if we don't **actually** sell it, we still may ***not*** recognise this as a diminution or impairment of value. Instead we continue to value the asset at the historic cost. Also, if an asset is worth less because interest rates have risen and therefore the discount rate that should be applied to the expected cash flows has risen (this makes the asset worth less), we do not recognise this as impairment either. If, however, there has been some trigger event, such as a delayed payment of interest by a borrower or a declaration by a rating agency that there has been an event of default (in the case of a corporate loan), then it would have to be written down to its appropriate amortised cost.

Mark-to-market (MTM) and fair value

In contrast to the historic or amortised cost basis for valuation, mark-to-market (and fair value which is a version of MTM) uses market prices as the starting point for valuation. Pure MTM simply takes the price in a market or the price at which a similar asset has been sold as the basis for value. **Fair value** has been defined as the amount at which the asset could be bought or sold in a current transaction between willing parties **other than in a liquidation sale**. This latter over-ride – other than in a liquidation sale – is extremely important. The reason is that in times

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of market stress, assets sold in a market tend to sell for much less than their **fundamental value** which is defined as the value of expected cash flows discounted at the appropriate risk adjusted discount rate. This is known as the **discounted cash flow** valuation or DCF value. There is often a substantial discount on the DCF value as a result of market selling pressure.

Trading book and banking book

Having considered the possible rules for valuation – historic/ amortised cost or fair value – the next question is which of these two rules should be used for each class of asset on a bank's balance sheet. The regulators have decided that the appropriate rule depends on the **intention** of a bank with respect to its assets. If its intention is to **hold assets to maturity** there is one set of rules; if its intention is to **trade its assets** there is a different set of rules.

The regulatory rules therefore require banks to categorise their balance sheet assets into two classes:

- Assets intended to be held to maturity, such as loans, are allocated to the **banking book** and held there until maturity at the price paid for them unless, as we noted above, there has been some trigger event which requires us to write the asset down in value**.**
- Assets which the bank may have an intention to sell or trade and therefore does not intend to hold to maturity have to be allocated to the **trading book**, i.e. the part of the balance sheet for assets which may be sold in market transactions and have to be revalued to reflect market prices daily.

The relationship of market price to historical (book) cost for bank shares

When we consider the value of bank shares we can also compare with market price at which the bank's own shares trade in the market with the historical (balance sheet) value that the accountants put on the assets of the bank. This should give us some idea of how valuable investors think a bank's assets are compared with the view taken by the accountants working for the bank. If the price/book ratio is below one this suggests that investors don't believe that the accountant's valuations are realistic. If they are well above one, this suggests not only that investors think the asset valuations are appropriate i.e. not overvalued, but also that they believe the bank has a '**franchise value**' i.e. it can generate a return on its shares above the cost of its capital. I cover these ideas in more detail in Chapter 28.

Figure 1.2 shows price/ book ratios i.e. stock market price of the bank's shares/ historical cost valuation of the underlying assets for major banks up to early 2011. In good times the book value multiple as it is called, may rise to as high as four times. Since the crisis, few trade much above 1 and, in 2011, many were trading well below book with some in Europe even as low as 0.1. At the time of writing, Lloyds Bank was trading at around 0.33 and its price/earnings ratio on expected profits in 2012 was around 5.

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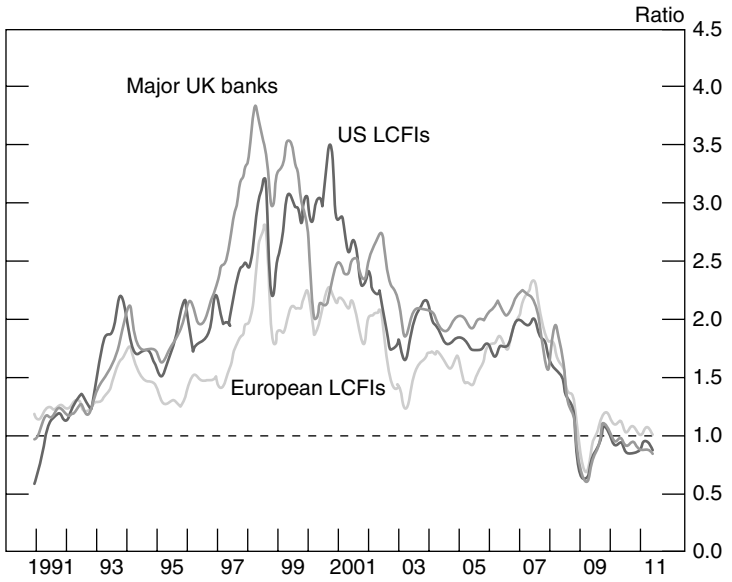


Figure 1.2 Major UK banks' and Large Complex Financial Institutions' price to book ratios

Sources: Bloomberg, DataStream and Bank of England calculations, Financial Stability Review June 2011 (p. 33), http://www.bankofengland.co.uk/publications/fsr/2011/fsr29sec3.pdf (accessed, 2 February 2012).

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